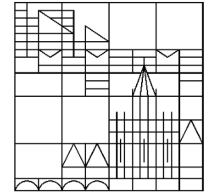


Universität
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Adult language acquisition and syntactic change

George Walkden

Universität Konstanz

AMC Symposium, 3rd December 2024

Slides: <http://walkden.space/AMC.pdf>



STARFISH

SOCIOLINGUISTIC TYPOLOGY
AND RESPONSIVE FEATURES
IN SYNTACTIC HISTORY

This talk



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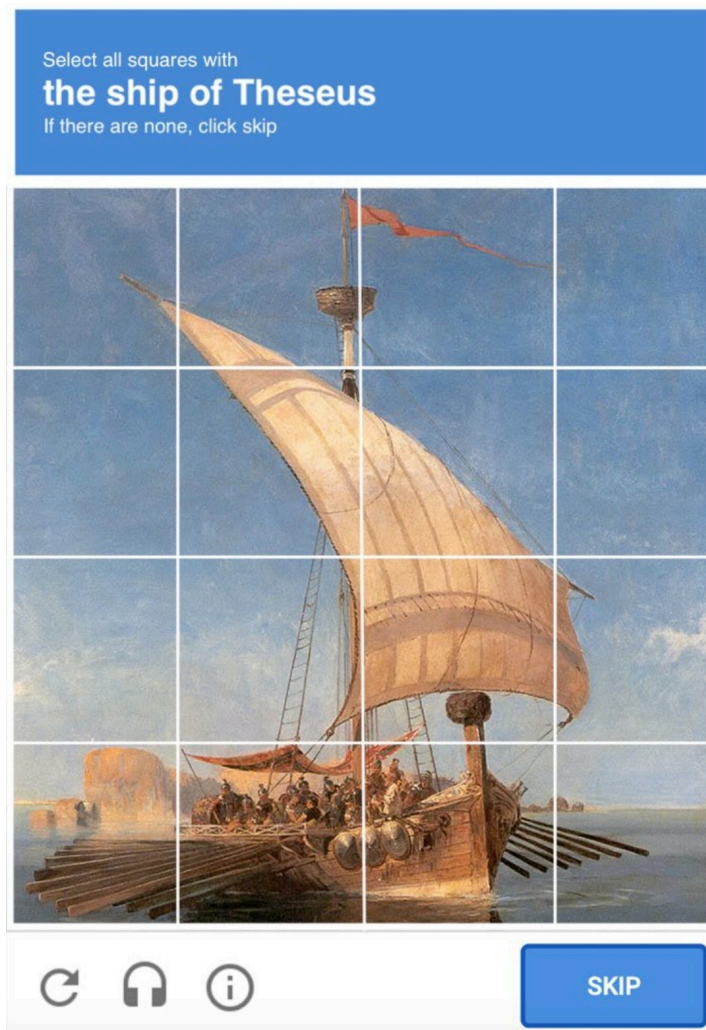
What is the place of adult language acquisition in theorizing about syntactic change?

- Child vs. adult language acquisition
- Sociolinguistic typology
- Conceptual foundations for adult language acquisition
- Studying adult language acquisition and change empirically: the STARFISH project

“Languages” as the Ship of Theseus



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“Writers seem to regard grammars as historically transmittable, as objects floating smoothly through time and space ... this is essentially a mystical view; grammars are discontinuous—created afresh by each language learner, who is influenced only by the data to be mastered and the theory of grammar restricting available hypotheses”
(Lightfoot 1979: 388)

“the Ship of Theseus is simply a case where our concepts just don’t give an answer ... the objects that we talk about are really objects of thought which are constructed by mental operations”
(Chomsky 2009)

<https://knowyourmeme.com/photos/2160540-captcha>

There are no “languages in contact”



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“what we mean by ‘languages in contact’ is ‘users of language in contact’ and to insist upon this is much more than a mere terminological quibble and has far from trivial consequences”

(McIntosh 1994: 137)

Talk of language-level ‘mechanisms’ is usually unhelpful (Walkden 2021)

Sociolinguistic typology: Trudgill's puzzle

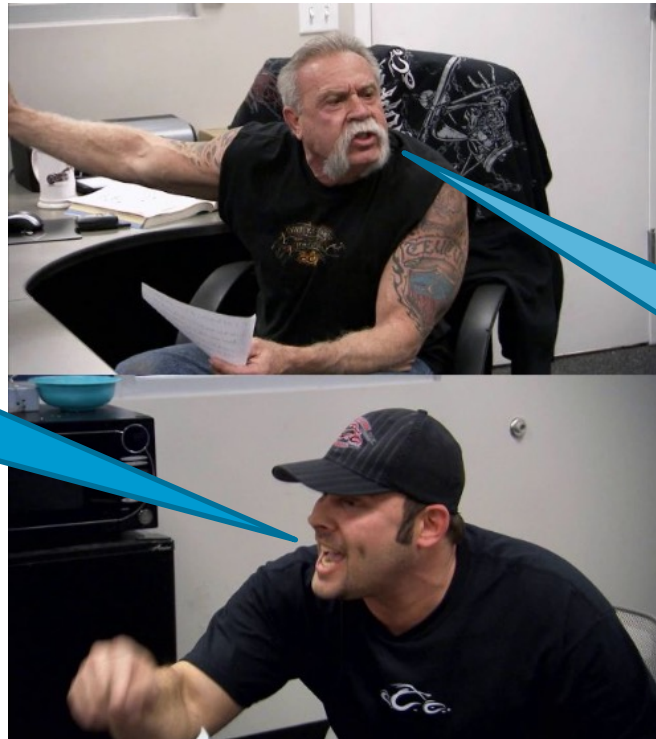


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From the call for this symposium: “What kinds of social conditions are likely to facilitate or impede different kinds of contact-induced change?”

contact leads to
simplification

Sociolinguists
(e.g. Trudgill 1986,
Milroy 1992)



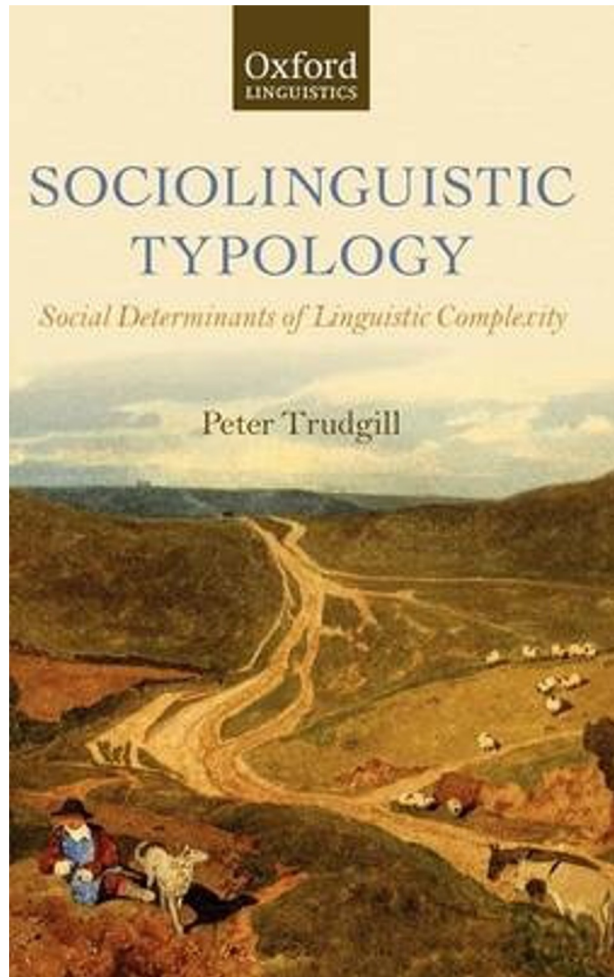
contact leads to
complexification

Typologists
(e.g. Nichols 1992,
Comrie 2008)

Sociolinguistic typology: Trudgill's solution



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“all the examples that support the claim that interference leads to simplification are of course counterexamples to the opposite claim”
(Thomason 2001: 65)

Trudgill (2011):

Different types of language contact situation may give rise to different types of change.

- short-term adult (L2+) language contact tends to lead to **simplification**
- long-term, co-territorial language contact tends to lead to **additive complexification**
- isolation tends to lead to **spontaneous complexification**

Defining simplification and complexification



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Simplification

- the **regularisation** of irregularities
- an increase in lexical and morphological **transparency**
- the loss of **redundancy** (syntagmatic and paradigmatic)

Example: loss of morphological distinctions in verb forms in Nubi Creole Arabic, spoken in Kenya & Uganda (Owens 1997, 2001)

Complexification

- irregularisation
- decrease in transparency
 - additional redundancy
 - can be **additive** or **spontaneous**

Additive: development of case marking and 5-way evidential system in Amazonian language Tariana under the influence of Tucano (Aikhenvald 2003)

Spontaneous: exaptation of –y ending for intransitive infinitives in dialects of the south-west of England (Ihalainen 1991)

The driving force: acquisition types



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*“it is very much a matter of **who does the learning, and under what circumstances**”*
(Trudgill 2011)



“while small children learn languages perfectly, the vast majority of adults do not, especially in untutored situations”

Some definitions



- **Child** language acquisition (L1+):
acquisition **during** the critical period
- **Adult** language acquisition (L2+):
acquisition **after** the critical period

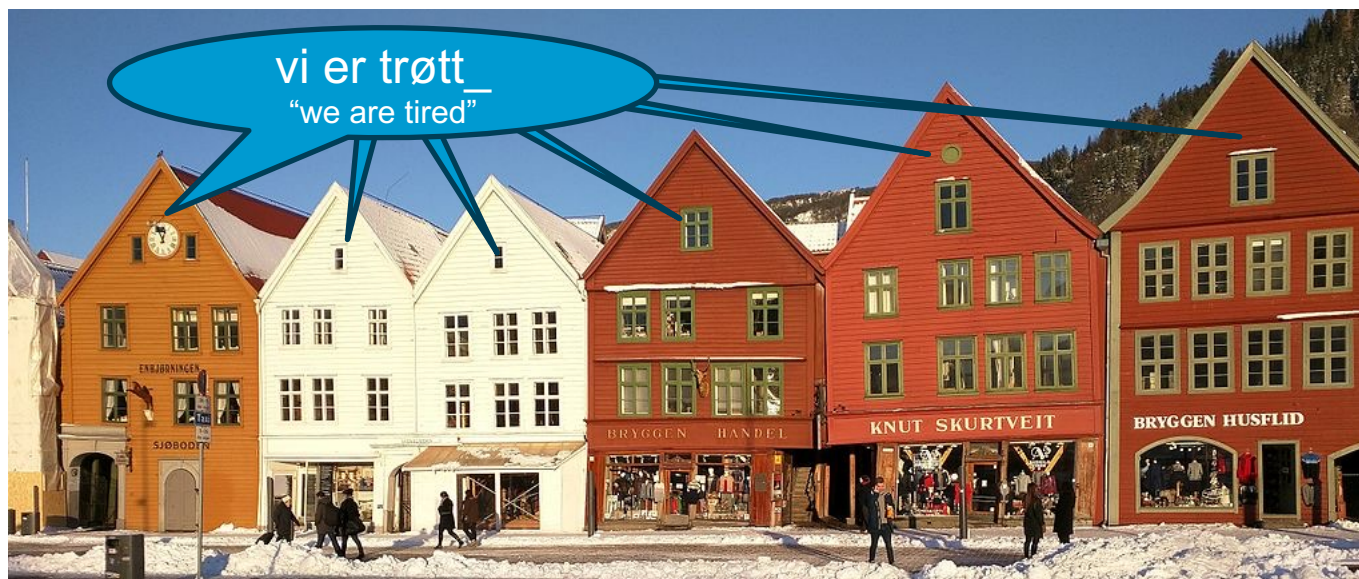
Existence of critical/sensitive periods is by now uncontroversial (Guasti 2016: §1.5)

In this talk I'll focus on adult language acquisition (causal factor in Trudgill's 'simplification').

At population level?



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Bryggen, by Greg Schechter (Wikimedia Commons, CC-BY 2.0)

Trudgill (2011: 57–58) on simplification of adjective inflection in Bergen Norwegian:

“My suggestion is that when the proportion of non-native becomes as close to 50% as this – the proportion of non-native speakers of Norwegian in Bergen was certainly very high (4,000 out of 9,000) – the number of face-to-face dialect-contact type interactions, and therefore potential instances of accommodation [...], would have reached a threshold level at which some aspects of the non-native variety could transfer to the native.”

What is this threshold? Why? How do we know? (cf. Kauhanen 2022; to appear)

Modelling the dynamics



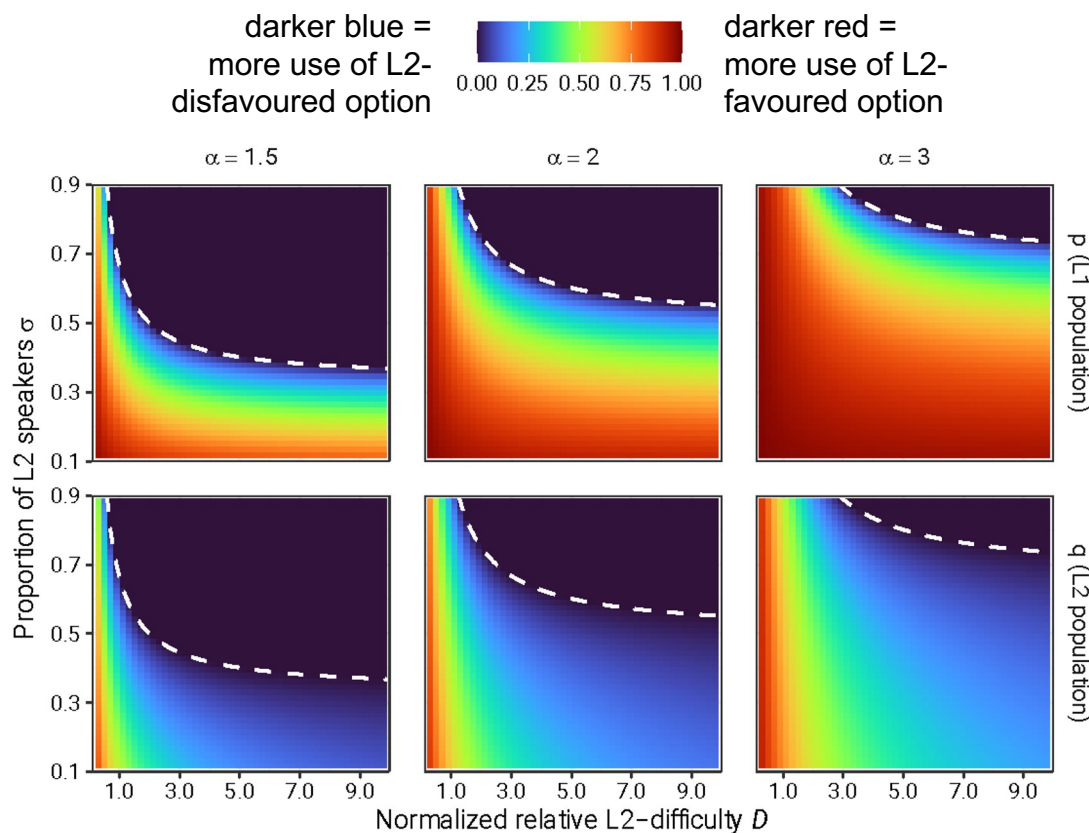
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How many adult learners do there need to be for them to have a diachronic effect?

Heat maps from Kauhanen (2022) showing interaction of L2-difficulty and proportion of L2 speakers in population.

Change away from L2-disfavoured features occurs when **L2-difficulty** (x-axis) and/or **proportion** of L2 users (y-axis) are high.

This will be dependent on the linguistic variable in question.



(Kauhanen 2022)

Transfer/CLI



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Transfer effects – what is usually called cross-linguistic influence (CLI) in the adult acquisition literature – are the most straightforward case of adult acquisition influence in diachrony.

Generally instances of “imposition” in the terminology of van Coetsem (1988, 2000) and Winford (2005).

Example: Pereltsvaig (2017), building on Santorini (1989), argues that embedded V2 in Yiddish arises from L1 Slavic speakers learning Yiddish as adults and imposing aspects of Slavic syntax.

I’ll put aside transfer effects today.

Adult (second/subsequent) language acquisition



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In generative syntactic work, the diachronic consequences of adult language acquisition have been under-researched (important exceptions: Weerman 1993, Meisel 2011).

The project STARFISH (Sociolinguistic Typology And Responsive Features In Syntactic History; 2020–2025) is an attempt to remedy this situation.

Like Weerman and Meisel, I'll argue in this talk that adult language acquisition can play an important role in change – alongside (not instead of!) child language acquisition.

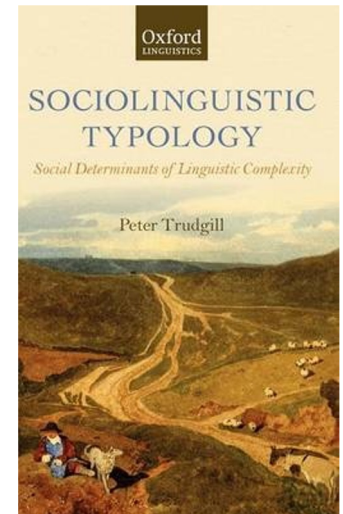
Investigating sociolinguistic typology empirically



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Walkden et al. (2023): we need to consider

- WHAT the linguistic innovation in question is
- WHO is in contact
- WHERE the contact took place
- WHEN the contact took place
- HOW the change happened
 - at the individual level
 - at the population level



Approaches for investigating sociolinguistic typology



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Before vs. after grammar comparison (Trudgill 2011)

- Pros: Straightforward; lots of data available
- Cons: May gloss over important details; potentially anecdotal

Typological-correlational studies (e.g. Lupyan & Dale 2010, Sinnemäki & Di Garbo 2018)

- Pros: Replicable; control for geography & relatedness
- Cons: Coarse-grained data; historical scenario may be underdetermined



Experimental studies

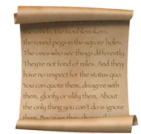
(e.g. Atkinson et al. 2018; Berdicevskis & Semenuks 2022)

- Pros: Allow for careful control and manipulation of individual-level factors
- Cons: May lack ecological validity

Historical corpus studies

(e.g. Blaxter 2017; Walkden et al. 2023)

- Pros: Access to the real-time historical causal factors
- Cons: Limited selection of languages available



Investigating sociolinguistic typology in the historical record



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No historically stable population consists entirely of adult language acquirers. Hence, establishing the historical facts of a situation is crucial to assessing the diachronic effects of adult language acquisition.

Investigating ‘orderly heterogeneity’ (Weinreich, Labov & Herzog 1968) in historical corpora.

- For instance, if a particular change in English is due to the influence of Norse adult learners, we would expect to see it earliest and most prominently in Middle English or late Old English texts, and in texts from north and east rather than from south and west.
- Need to consider factors such as register and text type (more on this later)

See Walkden et al. (2023) for discussion.

What's special about adult language acquisition?

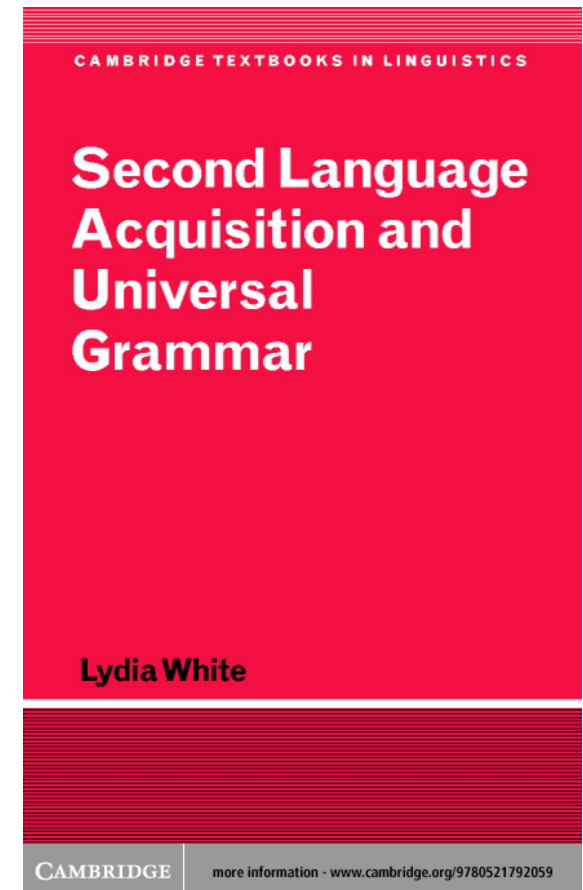


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In other words: what's the difference between acquisition within the critical/sensitive period(s) and acquisition afterwards, as regards syntax?

There is an entire field of scholarship dedicated to answering this question: see White (2003) for an overview.

Any attempt to theorize about the role of adult language acquisition in diachrony should take its lead from this literature, and not develop hypotheses *ex nihilo*.



UG or not UG: is that the question?



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Early (pre-2000) work attempted to characterize adult acquisition in terms of access (full, partial, or none) to UG.

In the context of GB/early P&P this made sense. But in the context of the Minimalist Program its usefulness is debatable (cf. White 2003: 17; Bley-Vroman 2009).

What is in UG? Biberauer (2019a):

- a. “the basic operations:
 - i. feature-sensitive – as opposed to ‘blind’ or Simplest – Merge, and
 - ii. likewise feature-sensitive Agree,
- b. a formal feature template of some kind (e.g. [iF]/[uF]), or possibly just the notion ‘formal feature, distinct from phonological and semantic feature’ (i.e. [F]) to be fleshed out in ways appropriate to the substantive content of the formal features in the system.”

Without all of this (and absent some other mechanism), no language is possible.

Is there a difference?



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Yes.

Bley-Vroman (2009): adult language learning is a) not reliable and b) not convergent.

Ultimate attainment is generally limited, if we use the right tasks to assess it.

Adult learners of English underuse syntactic information in processing wh-dependencies (e.g. Marinis et al. 2005), suggesting restricted access to certain structural information.

Even extremely proficient adult learners of Spanish (who perform at ceiling in offline comprehension tasks) display errors in elicited production of grammatical gender agreement (Grüter et al. 2012), suggesting different representations.

Are all differences due to transfer?



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No.

This view predicts that adult-acquired competence in aspects where the language acquired as a child and the language acquired as an adult are identical should also be identical (and performance should hence be at ceiling).

This prediction can be shown to be false.

- Sorace et al. (2009: 464): adult learners of any null subject language appear to “use overt subject pronouns as a compensatory ‘default’ strategy”, regardless of the structure of the language(s) they learned as a child.
- Bini (1993): this also holds for L1 Spanish speakers learning Italian.

Characterizing the difference



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Interpretability Hypothesis (Hawkins & Hattori 2006; Tsimpli & Dimitrakopoulou 2007)

Uninterpretable features are not accessible to adult acquirers.

In Minimalist syntactic theory, **uninterpretable** features:

- are present only within the syntax
- have no interpretation at the interfaces (i.e. no semantic content)

Applied to diachrony: Walkden & Breitbarth (2019), building on Trudgill (2011), predict that, in sociohistorical situations in which adult L2+ learners are dominant, uninterpretable features will typically be lost over time.

- STARFISH investigates this hypothesis (and related ideas).
- Quantitative analyses and fine-grained investigations of historical corpora (e.g. including geographical predictors) have the potential to shed more light on these questions than simple “before vs. after” grammar comparison.

“Patches” in adult language acquisition



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Yanovich (2019), in his commentary on Walkden & Breitbarth (2019), makes an important point: if Agree is driven by uninterpretable features, and uninterpretable features are unavailable to adult acquirers, then agreement should be completely impossible for adult acquirers, contrary to fact.

Proposal: in the absence of uninterpretable features, adult learners draw on general strategies of non-linguistic cognition in order to ‘patch’ their grammar

(the term is from Morgan 1972, applied to adult learning by Bley-Vroman 2009).

Patching using higher cognition plausibly incurs a higher processing and production cost.

This is one way to understand findings such as those of Grüter et al. (2012) on Spanish gender agreement in adult acquirers.

Two systems



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Kahneman (2011): human cognition is characterized by two modes.

System 1

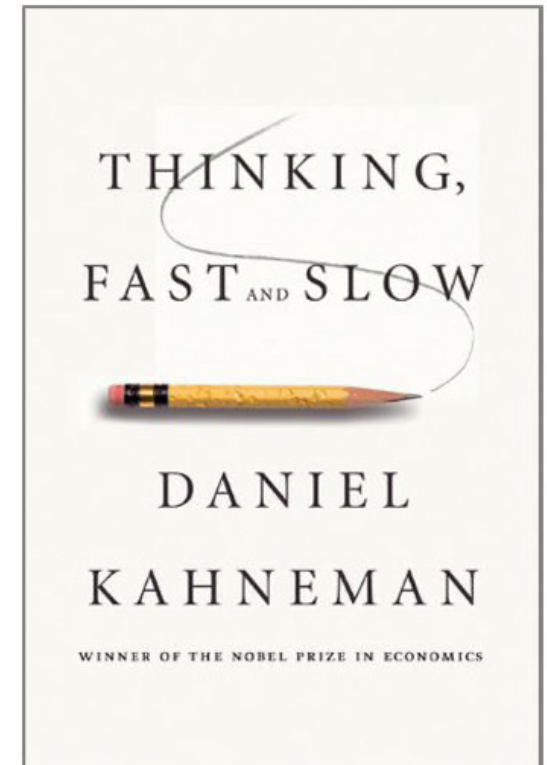
- fast
- instinctive
- below the level of consciousness
- characterized by heuristic reasoning

System 2

- slow
- effortful
- above the level of consciousness
- characterized by step-by-step reasoning

Theoretical linguistics provides, on the whole, a theory of (the linguistic module of) System 1: everything we know about child language acquisition suggests that it belongs here. Adult acquirers instead deploy System 2.

“Patches” are part of System 2.



Case studies



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Two case studies today:

- **Negation** (stage 2 > stage 3 of Jespersen's Cycle) (Walkden & Breitbarth 2019)
- **Null subjects** (McCarley 2024)

Other variables under investigation as part of the project:

- Gender and DP-internal phi-agreement
- Case
- Subjunctive mood

Syntax of negation



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Zeijlstra (2004 et seq.): bipartite negation (e.g. French *ne ... pas*) is syntactic agreement

Jespersen's Cycle:

- Stage I: *ne V*
- Stage II: *ne V ... pas*
- Stage III: *V ... pas*

Simplified representation (based loosely on van Gelderen 2011: 304):

- Stage I: $[\text{NegP } ___ [\text{Neg}' [\text{Neg}^0_{[\text{iNEG}]} \text{ne}] [\text{VP } \dots]]]$
- Stage II: $[\text{NegP } \text{pas}_{[\text{iNEG}]} [\text{Neg}' [\text{Neg}^0_{[\text{uNEG}]} \text{ne}] [\text{VP } \dots]]]$
- Stage III: $[\text{NegP } \text{pas}_{[\text{iNEG}]} [\text{Neg}' [\text{Neg}^0 \emptyset] [\text{VP } \dots]]]$

Under this analysis, the Interpretability Hypothesis predicts that Stage II is L2-difficult.

IMPORTANT: this doesn't imply that Jespersen's cycle is always caused by 'external' factors!

Case study: negation in Middle Low German

(Breitbarth 2014; Walkden & Breitbarth 2019)



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Transition from Stage II to Stage III of Jespersen's Cycle is simplification.

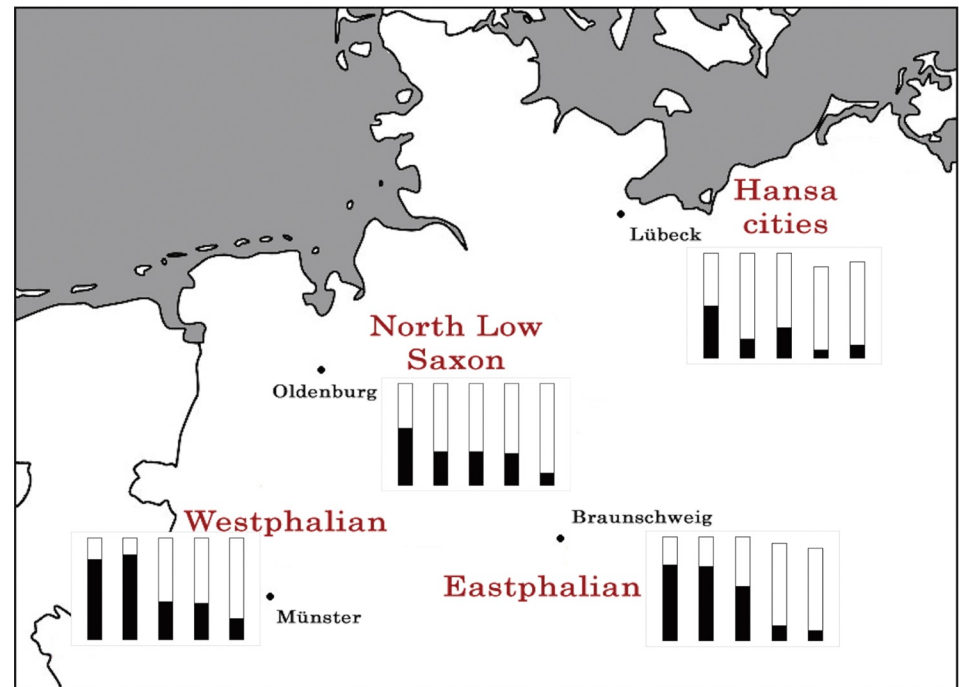
- Stage I: *en V*
- Stage II: *en V ... nicht*
- Stage III: *V ... nicht*

In Stage II, *en* is an agreement marker bearing an uninterpretable feature [*uNeg*].

dar **en** sculle wii se **nicht** ane hinderen
there NEG shall we them NEG from bar
“we shall not bar them from it”
(UB Lübeck 06/01/1450)

Hanseatic cities in the *Neuland* make the transition earliest and fastest (Breitbarth 2014).

Use of Stage II *en/ne ... nicht* (black) in the corpus of Breitbarth (2014)



Case study: null subjects in Latin American Spanish

(Walkden et al. 2023: §4; McCarley 2024)



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- **Spanish is a null subject language (NSL):**

Spanish [consistent NSL]:	<i>(Nosotros) queremos ir a la playa</i>
English [non-NSL (NNSL)]:	<i>*(We) want to go to the beach</i>

- **In Latin American Spanish (LAS) overt pronouns are being used at higher rates** (e.g. Dominican Spanish: Toribio 2000)
- **Moving through cycle toward becoming a NNSL?** (Camacho 2013)
- **Null subject licensing involves an Agree relation with a left-peripheral operator** (Frascarelli 2007), hence an extra uninterpretable feature
- **Null subjects are harder to acquire, particularly for adult acquirers** (Bini 1993, Pérez-Leroux & Glass 1999, Margaza & Bel 2006)

Syntax of null subjects



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- **Holmberg (2010: 94):**

“the property which consistent NSLs have, that partial NSLs [and non-NSLs–GW] do not have, is a D(efinite)-feature as part of the ϕ -feature make-up of finite T”
(cf. Holmberg 2005, Roberts 2010a, Walkden 2014: ch. 5)

This allows ϕ P pronouns to incorporate into T (following Roberts 2010b), and hence they are not spelled out as phrases.

- **The [uD] feature on T is uninterpretable** (Holmberg 2010: 97)
and is valued by agreement with an A-topic in the left periphery (Frascarelli 2007)
- **Under this analysis, the Interpretability Hypothesis predicts that consistent NSL-ness is L2-difficult.**

Null subjects in Latin American Spanish



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- Following Trudgill (2011), short-term, loose-knit, adult language learning → loss of L2-difficult features
 - Exact context for African learners of Spanish in colonial Latin America
 - These adult learners of L2+ Spanish might have struggled acquiring the L2-difficult null subject system, preferring overt pronouns
 - Their children would then have nativized this system
- This is exactly the scenario Sandro Sessarego (2013) proposes for Latin American Spanish where AHLAs (Afro-Hispanic Languages of the Americas) are these nativized varieties
 - They reflect the kind of change predicted: specifically overuse of overt subject

Yo no tengo plata, yo no quiere comprá.
I no have-1.SG money I no want-3.SG to buy
'I do not have money, I do not want to buy.' (Sessarego 2021: 107)

- Next step: look into the diachronic trajectory of pronoun realization in LAS

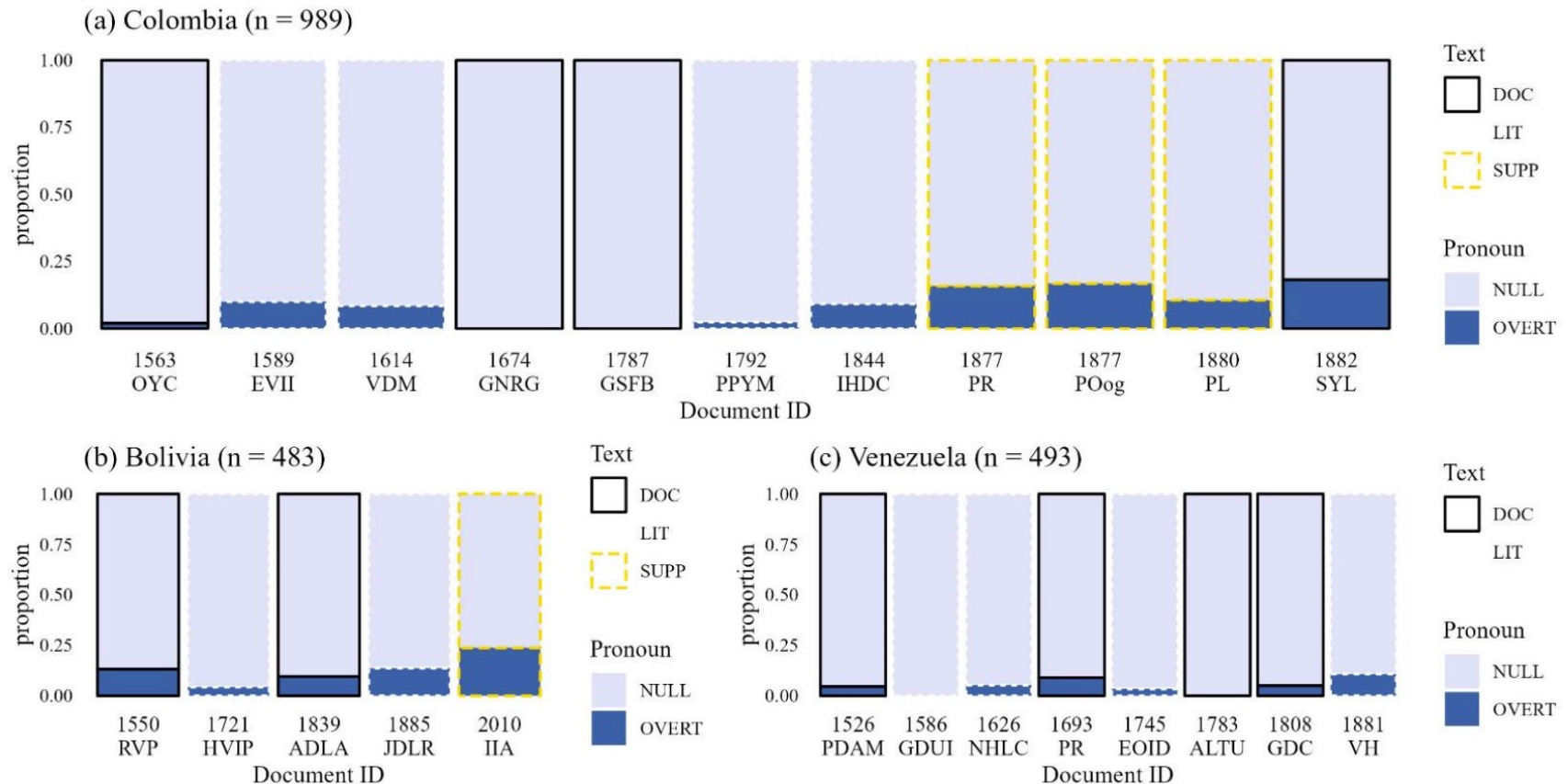
Corpus findings, C15th–19th

(McCarley 2024: ch. 6)

- Corpus: CorDELES; genres: literary, nonliterary, supplemental
- Mixed-effects regression with pronoun nullness as dependent variable
 - Significant predictors: orality; year



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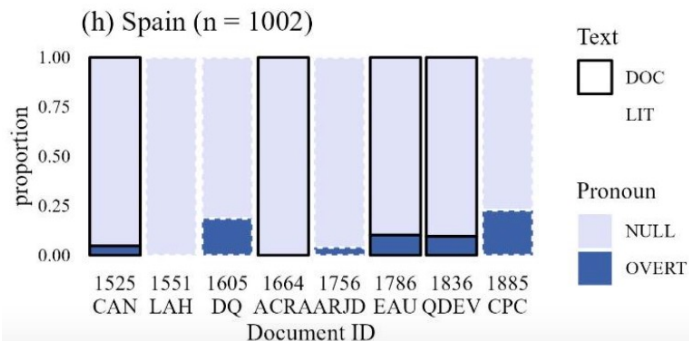
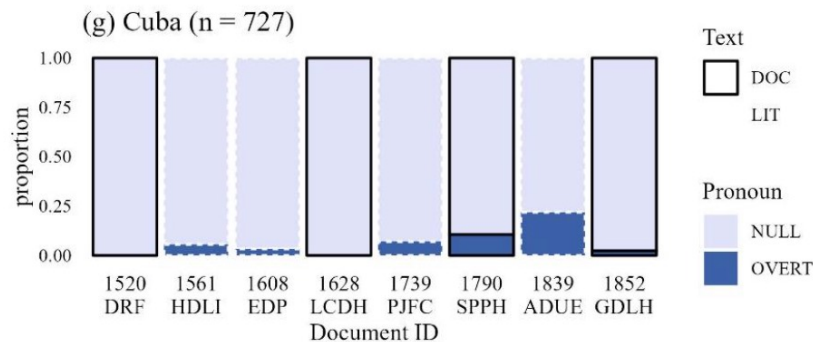
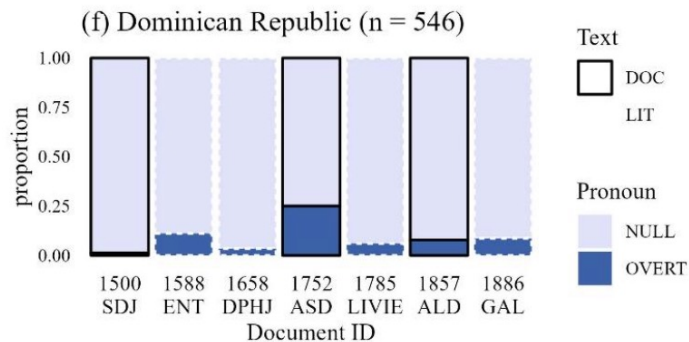
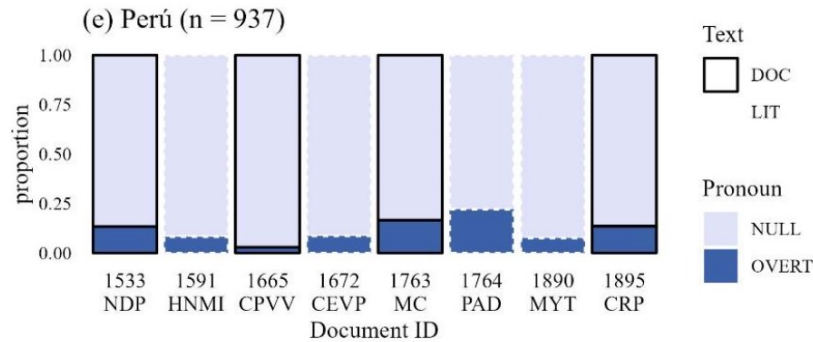
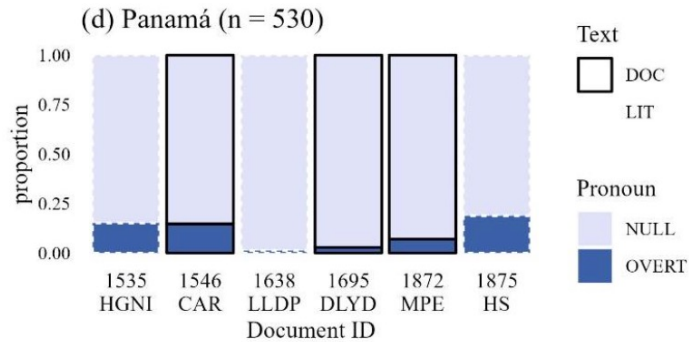


Corpus findings, C15th–19th

(McCarley 2024: ch. 6)



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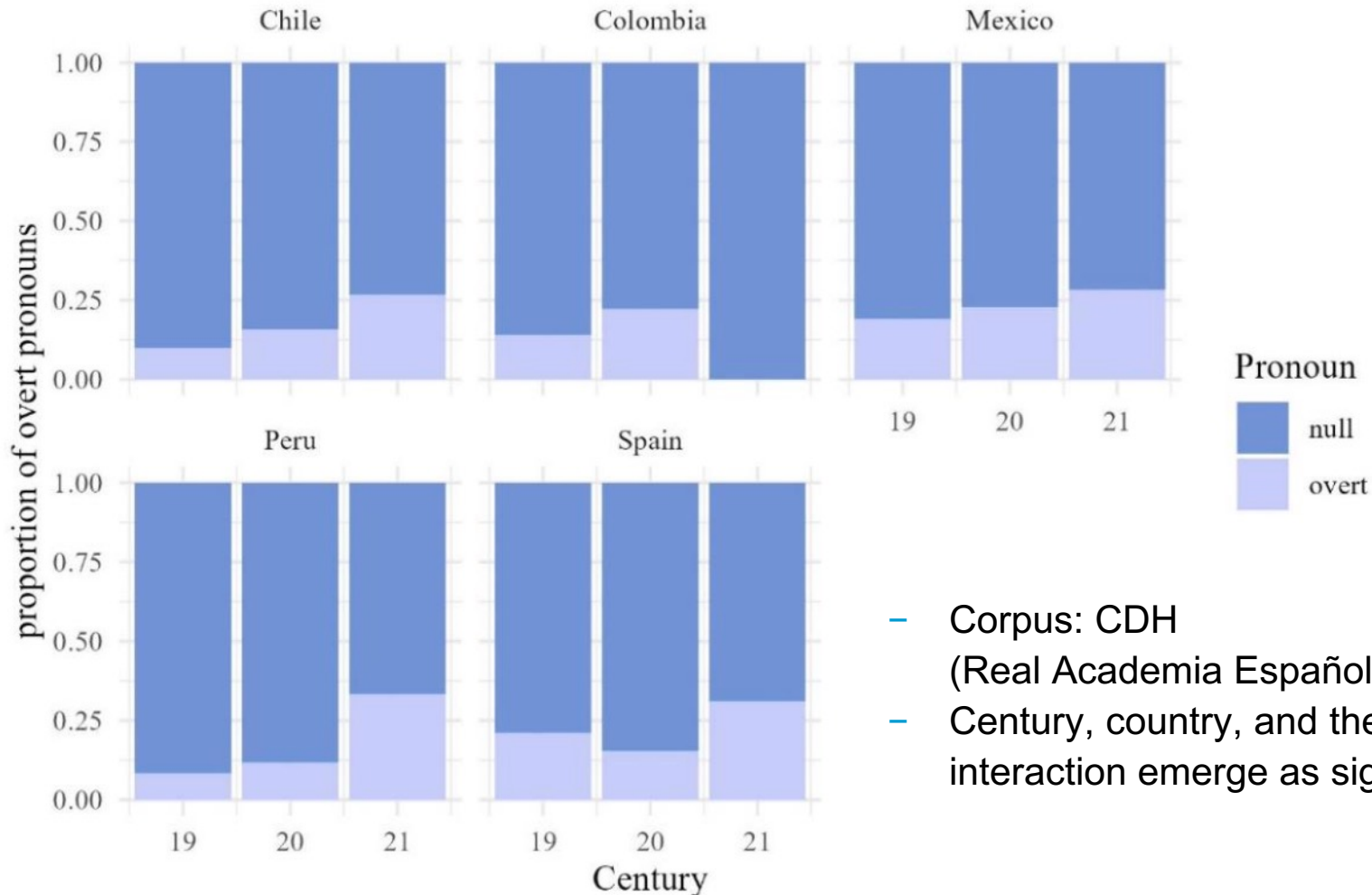
- Corpus: CorDELES
- Region/country not significant (!)

Corpus findings, C19th–C21st

(McCarley 2024: ch. 6)



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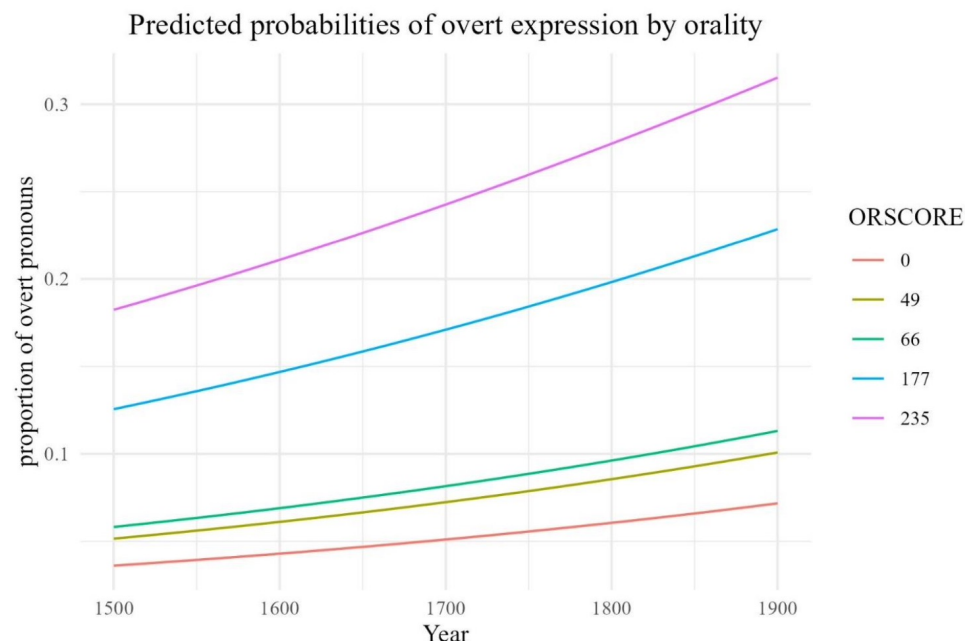
- Corpus: CDH (Real Academia Española 2013)
- Century, country, and their interaction emerge as significant

Null subjects in LAS: further questions



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- **An orality effect complicates the picture: more oral texts display more null subjects**
 - Orality is operationalized as a continuous measure using prototypically oral features (McCarley 2024)
- **Proportions of adult learners were generally high...**
 - Kauhanen (2022: 21), based on Bowser (1974): between 23% and 56% of the population of 17th-century Lima (Peru) were adult learners of Spanish
- **...so why aren't null subjects lost entirely?**
 - Under Kauhanen's model (building on Yang's Variational Learner), a very high threshold is predicted for the loss of null subjects



Conclusions



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Tentatively: there is some evidence from quantitative historical corpus studies that syntax is responsive to sociohistorical settings.

But more of this kind of work is needed!

In particular, good historical corpora are needed.

Thanks for your attention!

Thanks to the team: Henri Kauhanen,
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Molly Rolf and Sarah Einhaus

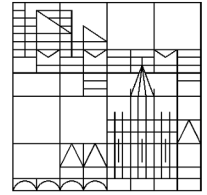
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<https://twitter.com/KonstanzLing>



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SOCIOLINGUISTIC TYPOLOGY
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